

NEPHROLOGY TEACHING SERIES

Hepatorenal Syndrome

AKI in the cirrhotic — a diagnosis of exclusion

Most AKI in cirrhosis is NOT HRS. Exclude prerenal, ATN, GN, obstruction first.

Dr. Cheng • Premier Nephrology Medical Group

MS3/MS4 rotation teaching

Why we get consulted

Rising Cr in a cirrhotic with ascites — primary team wants the HRS call.

CONSULT TRIGGERS

- AKI in cirrhotic with ascites
- Refractory ascites + rising Cr
- Diuretic hold + albumin — responded?
- Terlipressin vs midodrine/octreotide
- KRT: bridge or futile?
- Transplant / MELD trajectory
- SBP screen + treatment

TEACHING PEARL

HRS is never the first answer. Most AKI in cirrhosis is prerenal or ATN — HRS ~12%. Rule out everything else first. Paracentesis to exclude SBP is mandatory.

AKI in cirrhosis — the differential

Always walk through every cause before calling HRS.

PRE-RENAL (~44%)

GI bleed, over-diuresis, large-volume paracentesis without albumin, lactulose diarrhea, poor PO. Responds to volume/albumin.

ATN (~30%)

Sepsis (SBP, cellulitis, UTI), contrast, nephrotoxins. Does NOT respond to volume alone; muddy brown casts.

HRS (~12%)

Physiologic: splanchnic vasodilation → systemic hypotension → renal vasoconstriction. DIAGNOSIS OF EXCLUSION.

POST-RENAL

Uncommon. Tense ascites can rarely obstruct. Always get an ultrasound.

GN / INTRINSIC

IgA (alcoholic cirrhosis), HCV → cryo/MPGN, HBV → membranous. Check UA + serologies.

DRUG / TOXIC

NSAIDs (especially toxic in cirrhosis), aminoglycosides, vancomycin, ACE inhibitors in borderline.

HRS-AKI diagnostic criteria (ICA-ADQI 2024)

All criteria must be met — and alternatives excluded.

1

Cirrhosis with ascites (both required; HRS does not occur without ascites).

2

AKI by KDIGO: serum Cr $\uparrow \geq 0.3$ mg/dL within 48 h OR $\geq 50\%$ baseline rise within 7 days.

3

No improvement after adequate volume resuscitation when clinically indicated. ICA-ADQI 2024 recommends against mandatory 48-h albumin for everyone.

4

Absence of strong evidence for another primary cause: shock, nephrotoxin injury, obstruction, ATN, GN, or uncontrolled sepsis/SBP.

5

HRS-AKI can coexist with CKD, mild proteinuria, hematuria, or tubular injury — the key is whether another cause better explains the AKI.

SBP — the precipitant you must rule out

Diagnostic paracentesis on every cirrhotic with AKI. Treat before you search for HRS.

WHO NEEDS PARACENTESIS

Any cirrhotic with ascites AND any of:

- Admission with AKI or fever or abdominal pain
- Encephalopathy, GI bleed, or new leukocytosis
- Unexplained clinical deterioration

Diagnostic threshold: PMN > 250/μL in ascites

Send: cell count + diff, albumin (serum-ascites albumin gradient (SAAG)), culture in blood culture bottles

Do NOT wait for labs before empiric therapy if high suspicion

TREATMENT

Empiric: ceftriaxone 2 g IV daily × 5–7 days

Ceftazidime or carbapenem if nosocomial / recent hospitalization

Albumin: 1.5 g/kg on day 1, 1 g/kg on day 3 — ↓ HRS and mortality (Sort et al, NEJM 1999)

Hold non-selective BBs during episode (hypotension risk)

Stop diuretics if AKI present

Secondary prophylaxis: norfloxacin or SMX-TMP daily

HRS-AKI treatment

Vasoconstrictors + albumin bridge to transplant. KRT is supportive, not curative.

Option	Regimen	Setting
Terlipressin (Terlivaz)	0.85 mg IV q6h (CONFIRM dose); escalate to 1.7 mg q6h if Cr not ↓ ≥ 30% by day 4; up to 14 d	ICU/ward with monitoring; avoid if volume overloaded (respiratory failure ~11% vs ~2% in CONFIRM)
Norepinephrine (ICU)	0.5–3 mg/h IV to target MAP rise by 10 mmHg	ICU-only; alternative where terlipressin unavailable
Midodrine + octreotide + albumin	Midodrine 7.5 mg PO TID (titrate to 12.5) + octreotide 100 µg SC TID + albumin 20–40 g/day	Floor/outpatient bridge — less effective than terlipressin
Albumin (adjunct)	1 g/kg on day 1 (max 100 g), then 20–40 g/day	Required along with any vasoconstrictor
KRT	Standard AEIOU indications	Bridge to transplant only; not restorative for HRS
TIPS	Selected refractory ascites / HRS candidates	Worsens encephalopathy; MELD must be favorable
Liver transplant	Definitive	Kidney often recovers; occasionally combined simultaneous liver-kidney transplant (SLK)

Prognosis and transplant candidacy

HRS is a sign of end-stage liver disease. MELD captures it; transplant is the answer.

PROGNOSIS

Without transplant: median survival weeks to months

Acute HRS (AKI): worse than non-acute HRS (CKD-like)

Need for KRT predicts high mortality without transplant

Terlipressin verified HRS reversal rate ~32% (CONFIRM primary endpoint); complete reversal improves survival

Early listing + MELD progression = priority allocation

Non-reversal after 14 days of terlipressin = treatment failure

SIMULTANEOUS LIVER-KIDNEY (SLK)

Considered when HRS prolonged — unlikely to recover kidney post-LT alone

UNOS SLK criteria (2017): any of—

- CKD: $\text{GFR} \leq 60$ for > 90 d PLUS dialysis or $\text{GFR} \leq 30$ at listing
- AKI on KRT for ≥ 6 weeks
- AKI with $\text{GFR}/\text{CrCl} \leq 25$ for ≥ 6 weeks
- Combination of KRT + GFR criteria totaling 6 weeks
- Metabolic disease (primary hyperoxaluria, atypical HUS)

Multidisciplinary review + transplant center decision

Landmark trials in HRS

What changed practice — and what's still unknown.

Trial	Year	Takeaway
Sort et al.	1999	Albumin + ceftriaxone vs ceftriaxone alone in SBP — ↓ HRS, ↓ mortality.
OT-0401	2008	Terlipressin + albumin vs albumin — first positive US trial.
REVERSE	2016	Terlipressin ↑ HRS reversal vs placebo (~20% vs ~13%), not significant — no FDA approval.
CONFIRM	2021	Terlipressin + albumin vs placebo + albumin — verified HRS reversal 32% vs 17%. FDA approval 2022.
Cavallin et al.	2015	Terlipressin continuous infusion vs boluses — similar efficacy, fewer AEs.
MARS / Prometheus	2012	Artificial liver support — no survival benefit in HRS.

Common pitfalls

The errors that lead to missed SBP, missed alternative diagnoses, or terlipressin harm.

AVOID

- Calling HRS without paracentesis
- Stopping volume at 24 h
- Skipping renal US
- Terlipressin in volume overload
- Ignoring UA (RBC casts = GN)
- Trusting FeNa in cirrhosis
- Delay in transplant referral

DO INSTEAD

- Paracentesis on every cirrhotic AKI
- Albumin 1 g/kg + hold diuretics
- Renal US, UA, proteinuria
- Check volume before terlipressin
- Bland UA supports HRS
- Prerenal + ATN first; HRS third
- Notify transplant team day 1

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 58-year-old woman with alcohol-related cirrhosis (MELD 22), ascites on furosemide/spironolactone, admitted with decreasing UOP. BP 92/58, HR 98. Cr 2.4 (baseline 1.1). Ascites tap: PMN 80/ μ L, culture pending. UA: bland, < 200 mg/day protein. US: normal kidneys. Diuretics held; albumin given for suspected low effective arterial volume; Cr remains 2.3 after reassessment.

QUESTION: Does she meet HRS-AKI criteria, and what's the next step?

Clinical case — answer

Key teaching points from the case.

ANSWER

Yes — likely HRS-AKI after exclusion workup (cirrhosis + ascites, KDIGO AKI, no response after indicated resuscitation/diuretic hold, bland UA, no obstruction, no shock, no clear alternative cause). Start terlipressin + albumin with volume monitoring, and notify transplant hepatology now.

TEACHING

HRS-AKI is a diagnosis of exclusion, but the 2024 ICA-ADQI criteria no longer require a fixed 48-h albumin challenge or strict proteinuria/hematuria cutoffs. Paracentesis is still mandatory because SBP can precipitate AKI; PMN <250 makes SBP less likely while cultures are pending. Terlipressin 0.85 mg IV q6h (CONFIRM dose) + albumin; assess volume status carefully because respiratory failure risk rises if overloaded. Alternatives: norepinephrine in ICU or midodrine/octreotide when stronger options are unavailable. SLK is for prolonged kidney failure: sustained KRT ≥ 6 weeks or GFR/CrCl ≤ 25 for ≥ 6 weeks, or CKD criteria.

REFERENCES

Hepatorenal Syndrome

Guidelines

ICA-ADQI 2024 HRS-AKI Consensus
AASLD 2021 Ascites in Cirrhosis
EASL 2018 Clinical Practice Guidelines
United Network for Organ Sharing (UNOS) SLK Allocation Policy (2017)

Landmark Trials

Sort et al. — NEJM 1999 (SBP albumin)
OT-0401 — Gastroenterology 2008
REVERSE — Gastroenterology 2016
CONFIRM — NEJM 2021
Cavallin et al. — Hepatology 2015
MARS / Prometheus — Hepatology 2012

UpToDate (2026 access):

Hepatorenal syndrome in adults: Pathogenesis and diagnosis • Hepatorenal syndrome in adults: Treatment • Spontaneous bacterial peritonitis in adults • Liver transplantation in cirrhosis with AKI